MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026
IV SEMESTER - Discipline Specific Elective (DSE) - MG4DSEPHY202
EXPLORING THE COSMOS: OBSERVATIONS, CELESTIAL BODIES, AND COSMIC EVOLUTION

SCHEME of Valuation

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer

Answer any **10** questions. Each question carries **2** marks

1. What defines a Zodiac constellation? [U] / [CO1]
   A zodiac constellation is defined as one of the 12 constellations that lie along the ecliptic, the apparent path of the Sun on the celestial sphere. *- 2 marks*

2. What is the primary cause of Earth's seasons? [U] / [CO1]
   The rotation axis of Earth is 23.5 degrees. Earth is revolving around the Sun with this tilt and therefore the amount of sunlight falling on different regions of the Earth varies timely. *- 2 marks*

3. Write the electromagnetic spectrum in the increasing order of frequency. [U] / [CO2]
   Radio waves, microwaves, infrared, visible light, ultraviolet, X-rays, and gamma rays *- 2 marks*

4. What happens to the parallax angle as the distance to a star increases? [K] / [CO2]
   Parallax angle decreases *- 2 marks*

5. Define focal ratio of a telescope and explain its physical significance. [U] / [CO3]
   The focal ratio of a telescope is the ratio of its focal length to its aperture diameter. *- 1 mark*
   It is an indication of the light gathering power and the speed of the telescope. The smaller the f-ratio, the more light gathering ability the telescope has or the faster the optic is and the brighter the target image will be for a specific exposure time. *- 1 mark*

6. What is a neutron star? [U] / [CO4]
   In the case of fairly massive stars, the core left after the supernova explosion consists of neutrons and neutrinos. In this case, the neutron degeneracy pressure will counter balance the inward gravitational attraction and thus the stable neutron star is formed whose density is of the order of 10^{17}kg/m^3 *- 2 marks*

7. Write the net reaction of the p–p chain. [U] / [CO4]
   The net reaction of the p–p chain is the fusion of four hydrogen nuclei (protons) into one helium-4 nucleus, releasing energy, two positrons and two neutrinos. *- 1 mark*

   $$4^1\mathrm{H} \rightarrow {}^4\mathrm{He} + 2e^+ + 2\nu_e$$ *- 1 mark*

8. Explain how the surface temperature of stars is related to colour. [An]/ [CO4]
   Stellar classification based on the surface temperature - *- 2 marks*
   Wien's law only *- 1 mark*

9. What are the possible final stages of most massive stars? [An]/ [CO4]
Neutron stars or Black holes - *2 marks*
Any one answer - 1 mark

10. Why the massive stars evolve rapidly and have less main sequence lifetime? [An]/ [CO4]
Because their intense gravity creates extreme core temperatures and pressures, causing them to burn through hydrogen fuel exponentially faster than lower-mass stars. - *2 marks*

11. What is an absorption spectrum? [U] / [CO4]
Produced when light passes through a cool gas, creating dark lines on a continuous spectrum - *1 mark*
Absorption happens when electrons absorb photons to jump to higher states, resulting in dark lines against a continuous background, revealing the element's composition. - *1 mark*

12. Draw the light curve of a typical Cepheid variable star. [U] / [CO5]
A typical Cepheid variable light curve exhibits a saw-tooth pattern with a rapid rise to maximum brightness followed by a slower, smoother decline, repeating periodically. Correct curve - 2 marks

13. What is cosmic microwave background radiation (CMBR)? [U] / [CO5]
The cosmic microwave background (CMBR) is the relic radiation, the earliest light in the universe, released roughly 380,000 years after its birth. - *1 mark*
It fills all space and is detected as microwave radiation with a temperature of 2.73K. - *1 mark*

14. What is Big Bang nucleosynthesis? [U] / [CO5]
Big Bang nucleosynthesis is the production of light atomic nuclei (primarily hydrogen, helium, and lithium) within the first few minutes of the universe's existence. - *2 marks*

[2x10 = 20]

Part B

Short Essay
Answer any **6** questions
Each question carries **5** marks

15. Give an account of the Equatorial coordinate system. In what scenarios is it more useful than the Ecliptic coordinate system? [U] / [CO1]
Based on the celestial equator and poles. Coordinates are RA & DEC - *1 mark*
Definition and brief explanation of RA & DEC with figures - *3 marks*
The Equatorial coordinate system is more useful than the Ecliptic coordinate system for scenarios involving the observation of stars, the construction of star catalogs and navigating based on the rotation of the Earth. *Any one advantage - 1 mark*

16. Distinguish between flux and luminosity. If a star has a luminosity of 3×10^{26} W and is located at a distance of 10pc, calculate the flux received at Earth. [A] / [CO2]
Luminosity is the total energy output (power) of a source per second. Flux is the apparent brightness or power per unit area measured by an observer, which decreases with the square of the distance. - *2 marks*
$L = 3 \times 10^{26}$ W, d = 10pc = 10 x 3.26 x 9.46 x 10^{15} m = 3.086 x 10^{17}
$F = L / (4 \pi d^2)$

$$F = \frac{3 \times 10^{26}}{4\pi \times (3.086 \times 10^{17})^2} \approx 2.507 \times 10^{-10}\ \text{W/m}^2$$

- 3 marks

17. Write a note on the antennas used in radio telescopes. [U] / [CO3]
A radio telescope is a specialized antenna and radio receiver used to detect radio waves from astronomical radio sources in the sky. Radio telescopes are typically large parabolic dish antennas. These dishes reflect incoming plane waves onto a focal point where a feed antenna collects the signals. A large metal dish, often constructed from mesh to reduce weight and wind load, focuses energy, allowing detection of extremely faint signals. *- 5 marks*

18. Describe the concept of event horizon in detail. [U] / [CO4]
Event horizon is the limit up to which we can communicate. It is the theoretical boundary surrounding a black hole beyond which the gravitational pull is so intense that escape velocity exceeds the speed of light. Even light cannot escape from this limit. *- 5 marks*

19. Explain any theoretical method to calculate the temperature of a star. [U] / [CO4]
Explanation based on Wien's Displacement Law (colour based method) or Stefan-Boltzmann Law (Luminosity-Based Method) or Spectral Line Analysis (Classification Method) *- 5 marks*

20. Compare spiral and elliptical galaxies using observable properties such as star population, gas content, and structure. [A] / [CO4]
Elliptical galaxies are egg shaped. They range from nearly perfect spheres, E0, to the flattest, E7. Elliptical galaxies seem to contain practically all old stars. They have little visible gas and dust, but infrared and X-ray observations reveal some. *- 2 marks*
Spiral galaxies have a disk with central bulge and spiral arms. They are divided into two subcategories. Normal spiral galaxies with a bright disk where spiral arms wind out from a bulging nucleus. For the Barred spiral galaxies the spiral arms unwind from the ends of a bar-shaped concentration of material. Spiral galaxies have large amounts of gas and dust in the disk and contain young, middle-aged, and old stars. *- 3 marks*

21. The Hα spectral line (emitted wavelength = 656.3 nm) from a galaxy is observed at 670.2 nm.
Estimate the redshift and recessional velocity of the galaxy. [A] / [CO5]

$$z = \frac{\lambda_{obs} - \lambda_{emit}}{\lambda_{emit}} = \frac{670.2 - 656.3}{656.3} = \frac{13.9}{656.3} \approx 0.021179$$

- 3 marks

recessional velocity v = z x c = 0.021179 x 3 x 10^8 = 0.063537 x 10^8 m/sec = 6353.7km/sec

- 2 marks

22. The apparent magnitude of the galaxy NGC3159 is 14.4, and recessional velocity with respect to the Milky Way 6940 km/s. Find the distance and absolute magnitude of the galaxy. (Given Ho = 70 km/s/Mpc) [A] / [CO5]

Using Hubble's law distance D = v/Ho = 6940/70 = 99.14 Mpc *- 2 marks*

Distance modulus m - M = 5 log(D/10)
ie, 14.4 - M = 5 log (99.14 x 10^6/10) =34.98
and M = -20.58 *- 3 marks*

[5x6 = 30]